Answers (for marking)

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1. (a) bottom = adj, vertical = opp, sloping = hyp. (b) left vertical = adj, top = opp, sloping = hyp. (c) θ at the top: left vertical = adj, bottom = opp, sloping = hyp. *The hyp is always opposite the right angle; the adj touches θ.*
2. (a) sin A = 3/5, cos A = 4/5, tan A = 3/4. (b) sin B = 4/5, cos B = 3/5. (c) sin A = cos B and cos A = sin B, **because** the side opposite A is the side next to B (A and B add up to 90°).
3. (a) x = 21 cm (35 is 7 × 5, so the opposite is 7 × 3). (b) Same angles means same shape, **so** every side is scaled by the same amount (the ratio opp ÷ hyp stays 3/5).
4. (a) … the opposite side is 0.574 of (about 57% of) the hypotenuse. (b) 20 × sin 35° = 11.5 cm. (c) 8.61 ÷ sin 35° = 15.0 cm.
5. (a) No: sin = opp ÷ hyp and the hyp is always the longest side, so the fraction is less than 1. (b) Yes, when opp > adj, e.g. any angle above 45° (tan 60° ≈ 1.73). (c) 45°: tan = 1 means opp = adj, so the triangle is isosceles and the two other angles are equal: (180 − 90) ÷ 2 = 45°.

Page 2 · Sides and angles

6. (a) cos (adj and hyp). (b) tan (opp and adj). (c) sin (opp and hyp).
7. (a) x = 12 sin 40° = 7.7 cm. (b) x = 9 tan 28° = 4.8 cm. (c) sin 35° = 6/x, so x = 6 ÷ sin 35° = 10.5 cm (check: longer than 6 ✓).
8. (a) tan θ = 5/8, θ = 32.0°. (b) cos θ = 7/10, θ = 45.6°.
9. 24 tan 32° = 15.0 m above eye level, + 1.5 = **16.5 m**. (Common slip: forgetting the 1.5 m.)
10. (a) A steep 5 m ladder (70°) must reach most of the way up, nearly 5 m; 1.71 m is far too short. (b) The height is *opposite* the 70° angle, so it needs sin, not cos (cos gives the distance from the wall). (c) 5 sin 70° = 4.70 m.

Page 3 · The circle and the wave

11. sin: 0, 1, 0, −1, 0. cos: 1, 0, −1, 0, 1. E.g. at 90° the point is at the very top: 0 across, 1 up.
12. (a) 150°: the point at 150° is the mirror image of 30° in the vertical axis, the same height. (b) Negative: 200° is below the horizontal axis. (c) Any angle between 90° and 270°, e.g. 120°; both negative between 180° and 270°, e.g. 225°.
13. (a) $\cos^2\theta = 1 - 0.64 = 0.36$, so cos θ = 0.6 or −0.6 (two points on the circle at height 0.8, one right, one left). (b) adj 5, hyp 13, so opp = $\sqrt{(169 - 25)} = 12$. sin A = 12/13, tan A = 12/5.
14. (a) Wave through (0,0), up to 1 at 90°, (180,0), down to −1 at 270°, (360,0). (b) Same shape, everything 1 higher: from 0 up to 2 and back to 0. (c) It moved the graph up by 1 (a translation).
15. (a) Centre 20 m, radius 15 m. (b) sin is at most 1 and at least −1, so 35 m and 5 m. (c) sin 210° = −0.5 (mirror of 30° below the axis), so 20 − 7.5 = 12.5 m. (d) 15 sin θ = 7.5, sin θ = 0.5: θ = 30° and 150°.

Page 4 · Challenge

16. (a) $h^2 + 1^2 = 2^2$, $h^2 = 3$, $h = \sqrt{3}$. (b) sin 30° = 1/2, cos 30° = $\sqrt{3}/2$, tan 30° = $1/\sqrt{3}$; sin 60° = $\sqrt{3}/2$, cos 60° = 1/2, tan 60° = $\sqrt{3}$. (c) tan 45° = 1; diagonal = $\sqrt{2}$, so sin 45° = $1/\sqrt{2}$.
17. **AQA mark scheme:**

    $\sqrt{3}/2 \times \sqrt{3} + 1/2 = 3/2 + 1/2 = 2$ **B3**
    B2: $\sqrt{3}/2 \times \sqrt{3} + 1/2$ · B1: cos 30° = $\sqrt{3}/2$ or tan 60° = $\sqrt{3}$ or sin 30° = 1/2.
    For B3 all steps must be shown. Allow equivalent expressions for all trig values.

18. **AQA mark scheme** (one of two methods):

    Method 1: sin 30 = 1/2 or tan 30 = $1/\sqrt{3}$ (or $\sqrt{3}/3$) or cos 30 = $\sqrt{3}/2$ **M1** · 4 × 1/2 × $1/\sqrt{3}$ × $\sqrt{3}/2$ **M1dep** · 1 with all three values seen **A1** · 90 **A1**
    Method 2: 4 × sin 30° × (sin 30° ÷ cos 30°) × cos 30° **M1** · 4 × 1/2 × 1/2 **M1dep** · 1 **A1** · 90 **A1**
    Accept any angle of the form 90 + 360n, where n is an integer.

19. (a) BD = 12 tan 35° = 8.40 m. (b) sin 60° = BD/BC, BC = 8.402… ÷ sin 60° = 9.70 m. (c) 8 ÷ sin 60° = 9.24 m: almost half a metre out. Rounding early throws away accuracy, and the error grows in every later step.
20. No. sin 40° = 0.643 is a **ratio**: the opposite side is 0.643 *times* the hypotenuse. It is only 0.643 cm if the hypotenuse is exactly 1 cm.